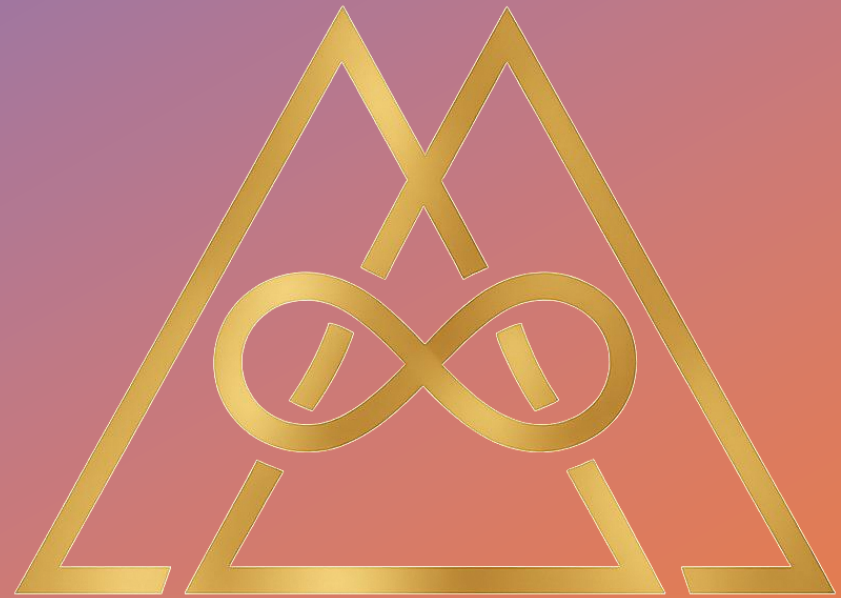


Martingale Protocol

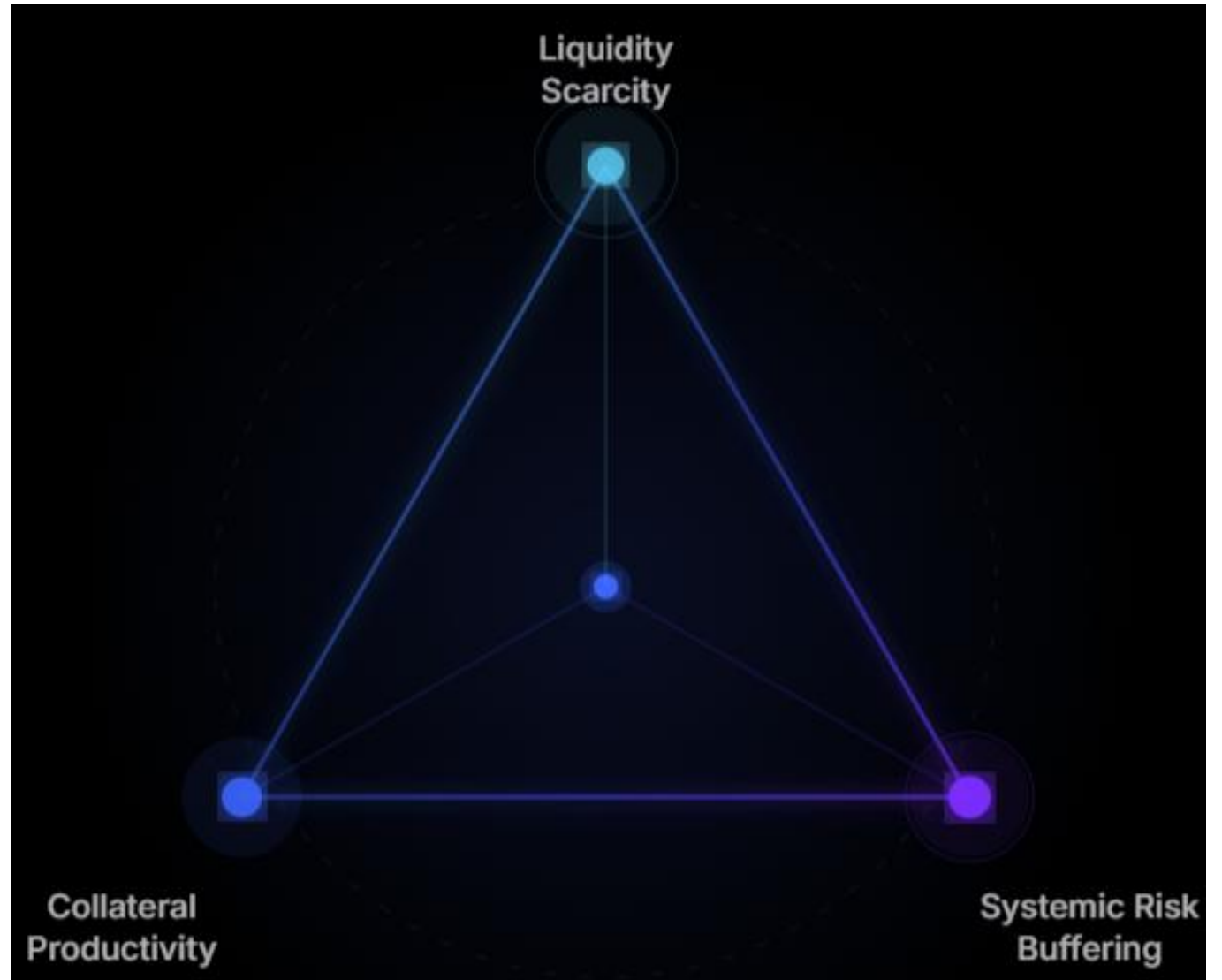


Martingale

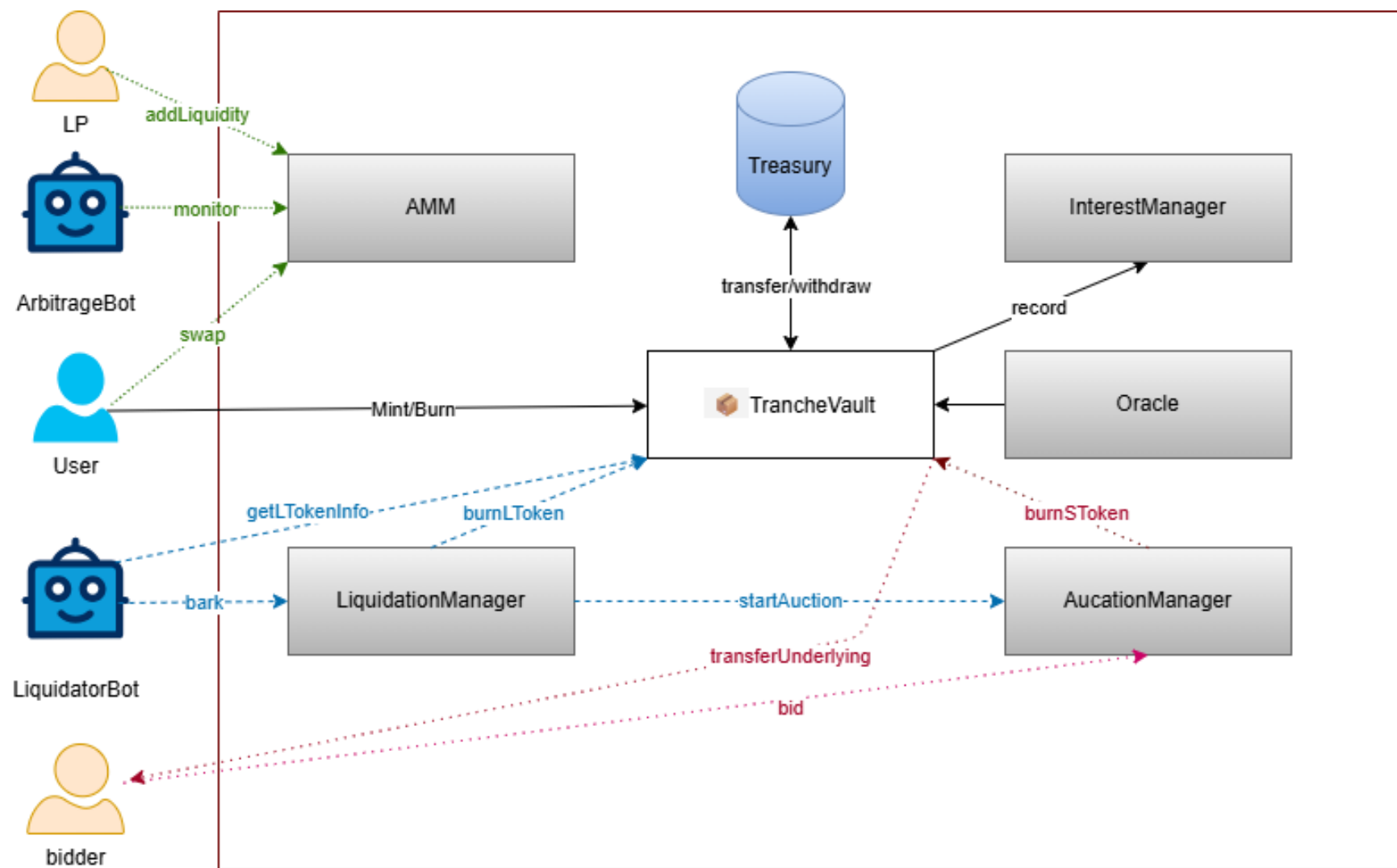


Overview

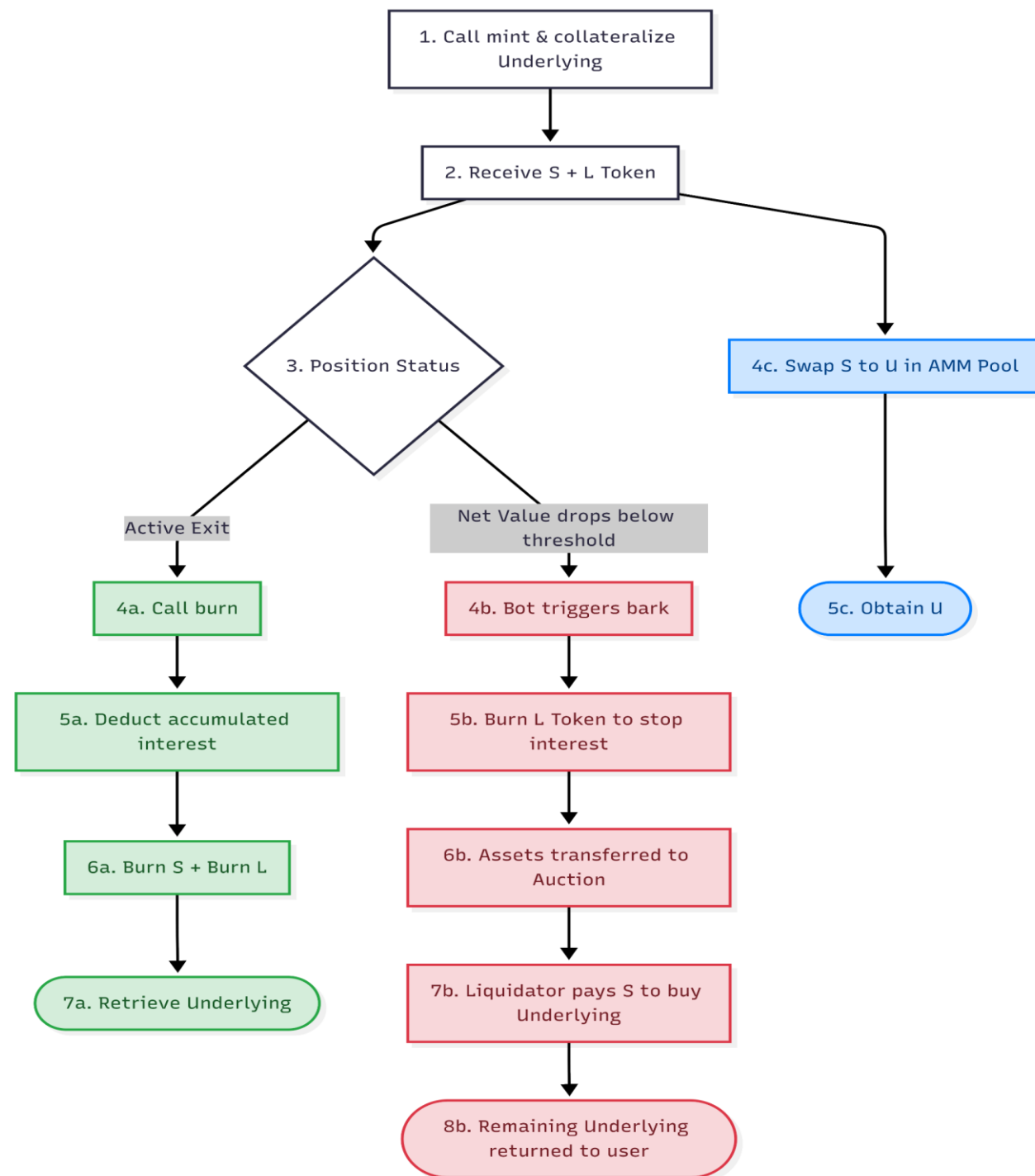
- Martingale protocol is **decentralized credit**, which combines stablecoin and lending Markets.
 - ✓ *Pledge underlying to borrow stable coins*
 - ✓ *Customized DeFi investment tools for users with different risk preferences*



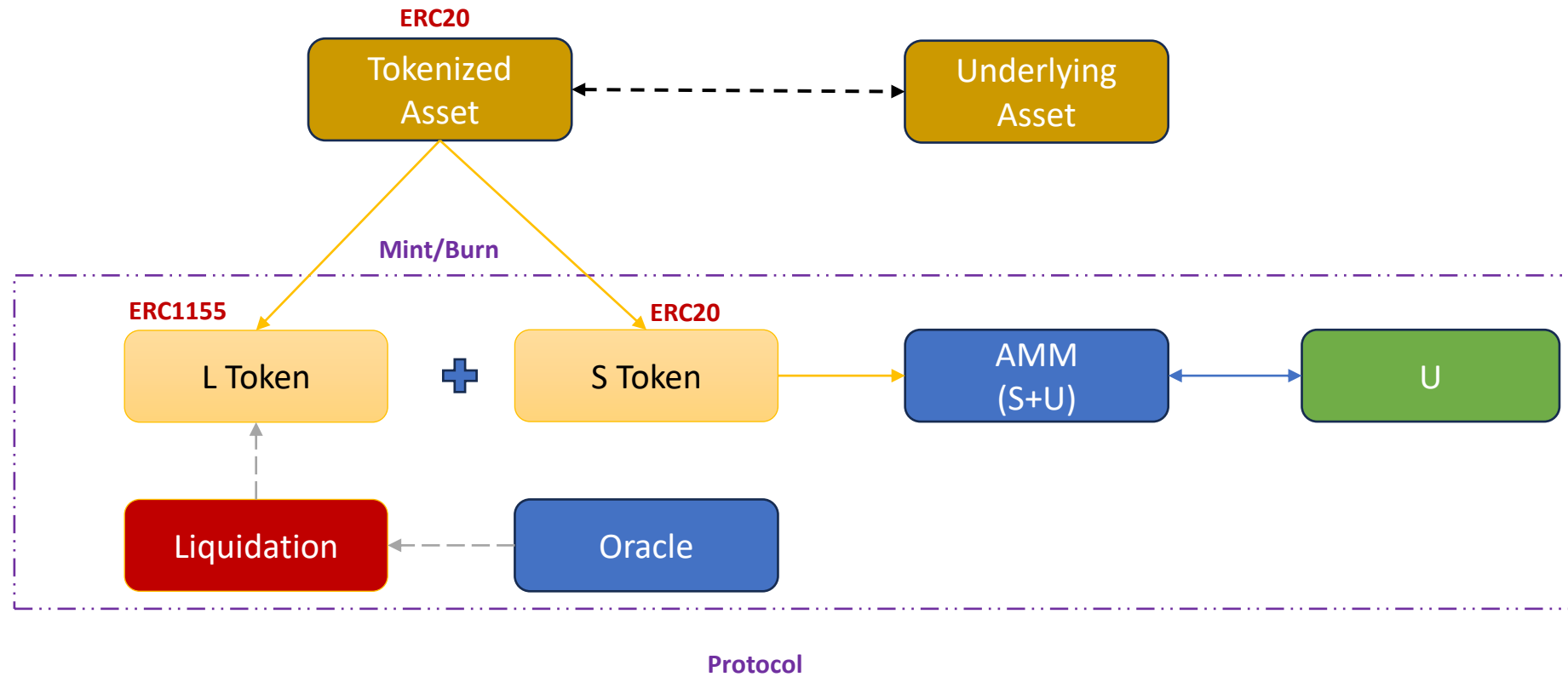
System Architecture



Life Cycle



Model I — — Mint/Burn



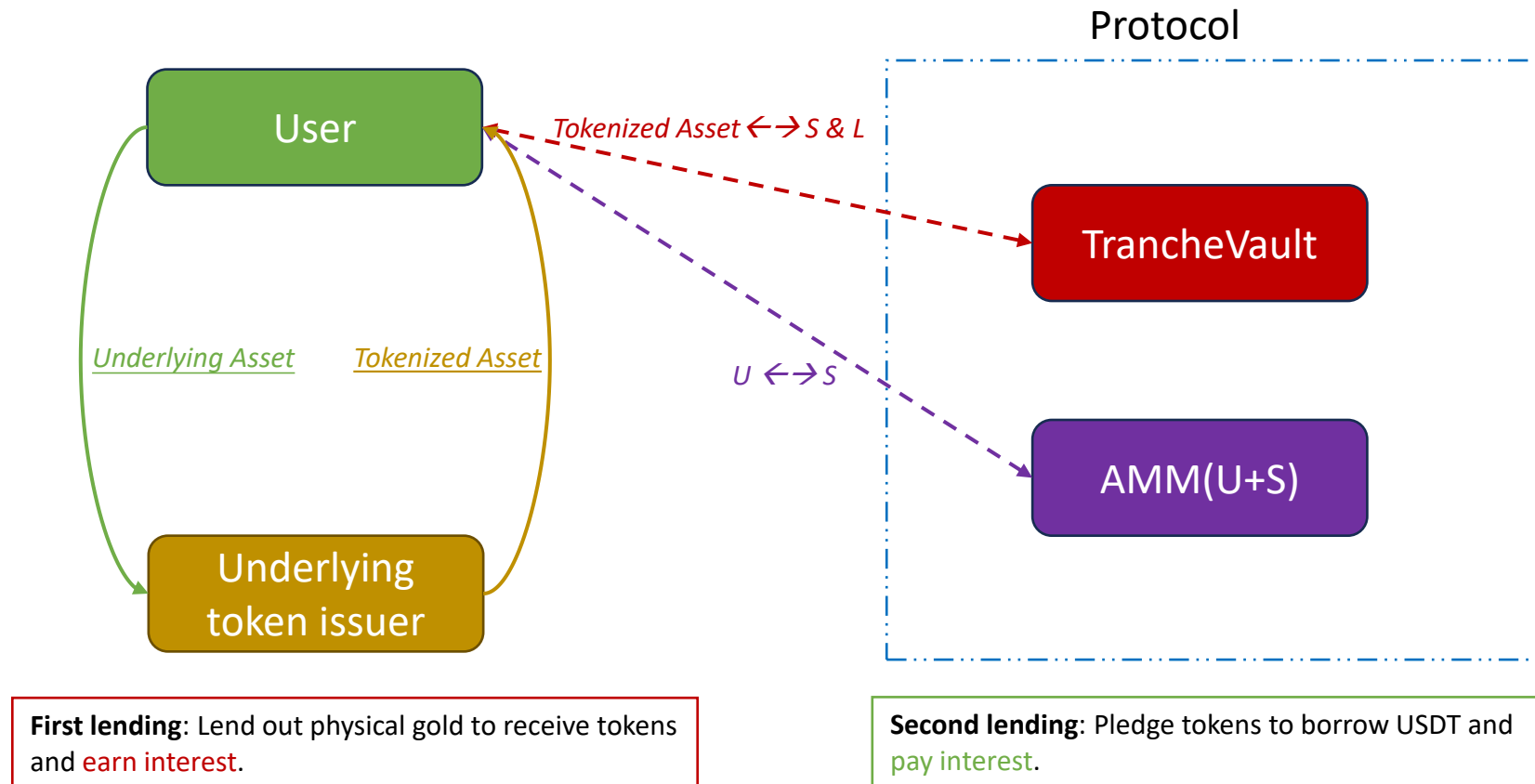
Model I — — Mint/Burn

- **Parameters:** $LTV_{\max} = 75\%$, $Liquidation = LTV_{\max} + x\%$
- **Mint:** Stake N_G Underlying $\rightarrow$ Mint N_S stable token + N_L leverage token

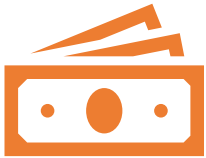
t = 0	t > 0
$N_S * 1 + N_L * 1 = N_G * P_0$ $LTV_0 = \frac{N_S}{N_G * P_0}$ $NAV_0 = 1$	$N_S * 1 + N_L * NAV_t = N_G * P_t$ $LTV_t = \frac{N_S}{N_G * P_t}$ $NAV_t = \frac{P_t - P_0 * LTV_0}{P_0 - P_0 * LTV_0}$

- **Burn:** Users can burn minted assets in any proportion to redeem the underlying assets.

Model II — Interest Rate Model



Model II — Interest Rate Model



Collateralize Tokenized Asset → Borrow U

Users pay interest to protocol



Model:

**Gross Borrow Rate = Base Rate +
Utilization Factor + Collateral Risk
Premium**

Base Rate: Lending interest rate for USD (U)

Utilization Factor: The fewer U in the AMM,
the higher the factor

Collateral Risk Premium: Risk premium of
the underlying asset



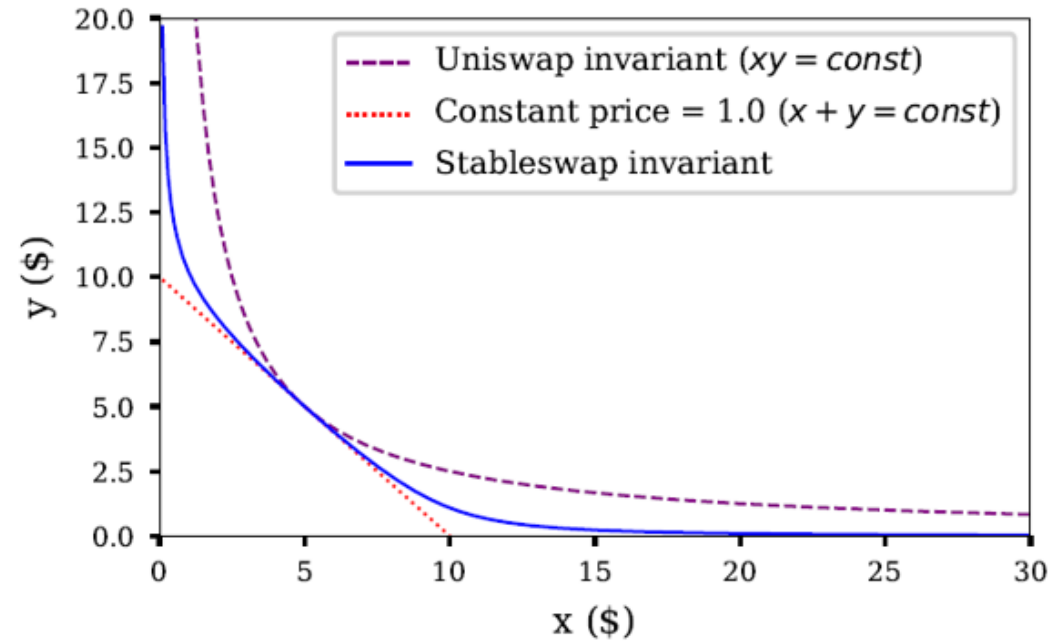
Practical:

Interest is accrued on a per-second basis
and reflected in the net value of L tokens.

Upon burning, the deduction is made once
in the form of the underlying asset

Model III — — AMM

- **AMM:**
 - S token + U
 - Stableswap invariant
- **Arbitrage:**
 - Use an arbitrage program to monitor AMM deviations.



Model IV — — Oracle



Third-party oracles

Chainlink

Pyth Network

RedStone



Core features:

Delayed price updates

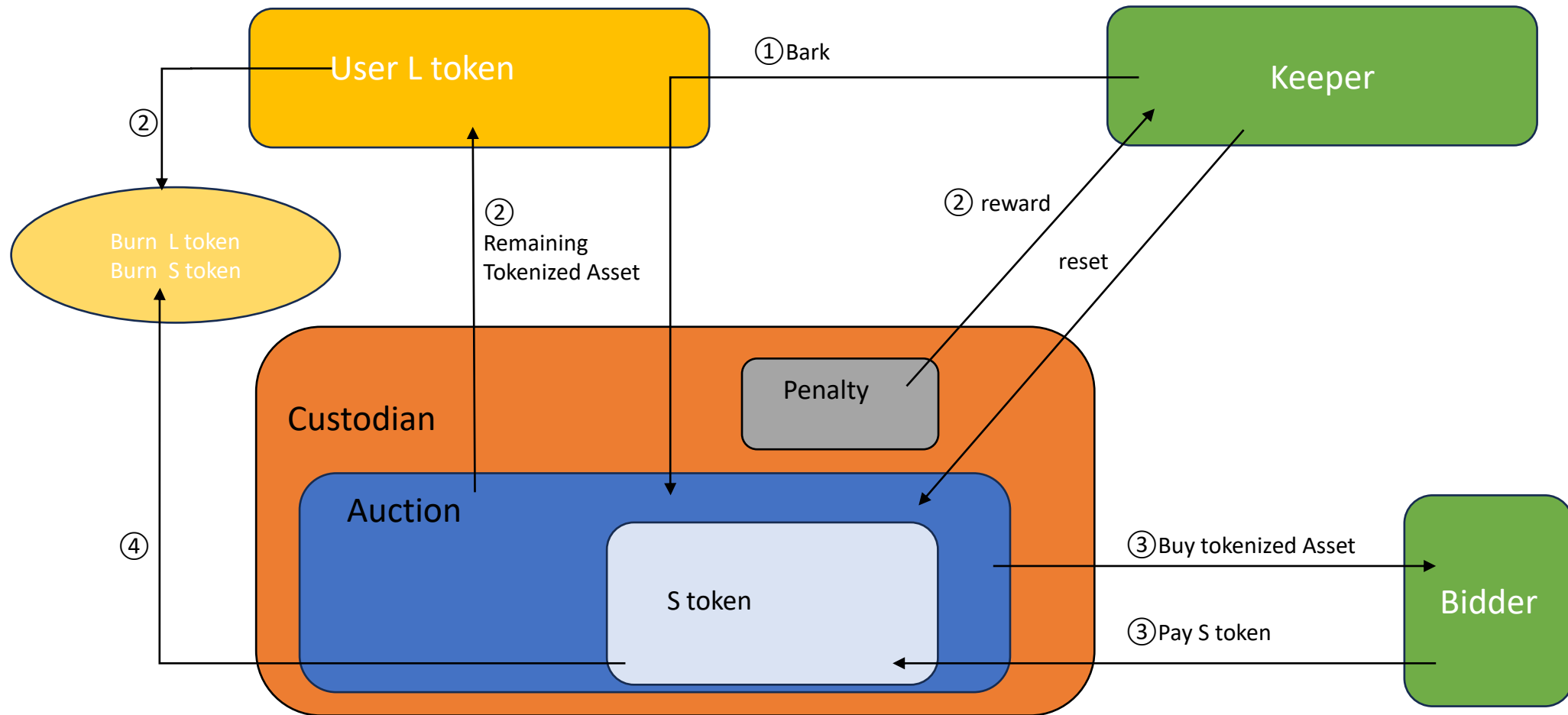
Multi-source aggregation

Multi-signature voting mechanism

Emergency halt

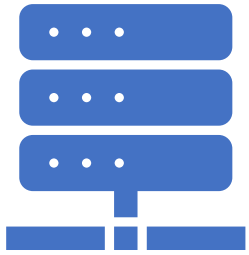
Price change threshold

Model V — — Liquidation



NOTE: The liquidation model determines the optimal liquidation boundary based on system parameters for users.

Mechanism



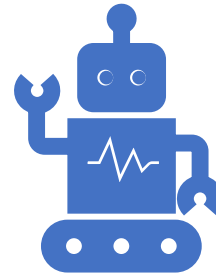
On-chain

Access Control / Multi-sig

Whitelist

Proxy Pattern

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Off-chain

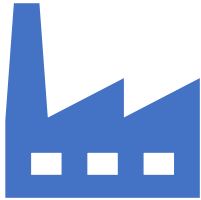
Liquidation bots

Arbitrage bots

Maintenances bots

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Website



Martingale: <https://martingaleco.com>



App Demo: <https://test-al2.pages.dev/app>



Documentation: https://test2026.gitbook.io/test-tranche_protocol/fMFG0viXW9hbW0MeCbeh/